Team 33: Investigating the Relationship Between Students' Lifestyle Choices and Academic Performance

Team Members: Chakri, Rutvik Sai, Yaswanth, Goutham Pinnani

October 12, 2026

1. The Main Hypothesis/Question

Main Question:

How does a student's daily time allocation across studying, sleeping, social engagement, and physical activity correlate with their academic performance and perceived stress levels? Can we identify statistically significant lifestyle patterns that distinguish high-performing students from their peers?

Basic Idea of the Research:

University life demands a delicate balance between academic responsibilities and personal well-being. A common challenge students face is managing their time effectively. This project aims to investigate the complex interplay between students' daily lifestyle choices—specifically time spent on studying, sleeping, social activities, and physical exercise—and their academic performance, often measured by GPA. We also seek to understand how these lifestyle factors contribute to or mitigate stress levels.

Using the "Student Lifestyle Dataset" from Kaggle, we will employ statistical methods to analyze these relationships. We hypothesize that while increased study time might positively correlate with higher academic performance, this relationship is not strictly linear and is heavily influenced by adequate sleep and moderate physical activity. Furthermore, we will test whether students who allocate sufficient time to sleep and social engagement exhibit lower stress levels, potentially indirectly benefiting their academic outcomes.

The study will focus on identifying trade-offs in time allocation. For example, does sacrificing sleep for extra study time yield better academic results, or does it lead to higher stress and diminished returns? By exploring these questions, we hope to identify optimal lifestyle patterns associated with higher academic performance that do not compromise well-being. This analysis is exploratory, and we acknowledge that external factors not captured in the dataset may also influence academic success and stress.

1.1 Data Set Source

- **Dataset Source:** Kaggle: Student Lifestyle Dataset.

1.2 Programming Languages or Tools/Software

- **Programming Language:** Python
- **Tools/Software:** Jupyter Notebook, pandas (data manipulation), NumPy, Matplotlib(visualization), SciPy (statistical testing), LaTeX (report generation), GitHub (version control).

1.3 Datasets

- **Primary Dataset:** Student Lifestyle Dataset (Kaggle) - Contains variables like Study Hours Per Day, Extracurricular Hours Per Day, Sleep Hours Per Day, Social Hours Per Day, Physical Activity Hours Per Day, GPA, and Stress Level.

1.4 Links to LLM Conversations

- https://chatgpt.com/share/6acd507f-4e98-83ec-816b-bb9d6468039b

2. Logistics

2.1 Updated Spreadsheet

Our project tracking spreadsheet will be updated regularly to document our dataset links, hypothesis formulations, and overall progress. The live document can be accessed at this link.

2.2 Proposed Breakup of Work

- **Chakri:** Inferences, Running hypothesis testing, Result interpretation and conclusion formulation
- **Rutvik Sai:** Plotting and visualization, Suggesting null and alternate hypotheses, Data Analysis
- **Yaswanth:** Parsing and cleaning data, Regression analysis, Feature selection
- **Goutham Pinnani:** Report work, Studying the distributions, Identifying correlation among the factors, Final compilation and presentation

Collaboration Plan: We plan to hold weekly sync-up meetings on Fridays in our Hostels. We will use a shared GitHub repository for code management.

3. Specific Help Needed

We would greatly appreciate your guidance in a few key areas as we proceed:

- Like determining when to prioritize correlation versus covariance for our specific variables.
- Choosing the most appropriate hypothesis tests based on our observed data distributions.
- Choosing optimal confidence intervals